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Full Length Article

Voluntary carbon market derivatives: Growth, innovation & usage

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Abstract

Voluntary carbon markets (VCMs) have emerged as a preferred market mechanism to support the global economy's transition to a low carbon future. Trading in carbon offset derivatives has moved into the financial mainstream with commodity exchanges now offering trading in VCM futures. This paper discusses the historical background of carbon trading, the market ecosystem of companies and organizations that provide the necessary infrastructure, and the evolution of VCM derivatives, focusing on CME Group listed instruments. The paper discusses findings that show that exchange-traded derivatives products can lead to increased standardization, transparency, and price discovery in nascent VCMs. We review academic literature published in this growing field of research and conclude by providing recommendations for areas of further market development and research.
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1. Introduction

The evolution of the voluntary carbon market (VCM) over the last two decades is a story of innovation, experimentation, mistakes, and corrections. The VCM, as its name suggests, is a voluntary, market-based mechanism for curbing global carbon emissions. This specialized market allows the private sector, governments, and individuals to participate in carbon offsetting and trading, helping entities offset their emissions while they work to decarbonize their supply chains and develop new, energy efficient technology. There is no single entity that manages the market, but rather an ecosystem of self-regulating standard-setting organizations and certifiers. In the last few years, the VCM has embraced financial derivatives. A number of platforms now offer trading in derivatives linked to VCM offset credits. This paper will discuss the emergence of futures contracts on the VCM and their role in increasing market efficiency. First, the paper will consider the historical development of carbon trading and the market ecosystem which underpins the physical market. Next, we will focus on the evolution of financial derivatives and how these products can play a meaningful role in helping the market manage commodity price risk. We will illustrate the growth of derivatives by focusing on CME Group's listed VCM contracts. In the next section, we review academic literature published on this subject. The following section will discuss ways in which this key market may develop and will conclude this paper.

2. History of the VCM

In 1997, the adoption of the Kyoto Protocol led to the establishment of global compliance carbon market mechanisms for the first time. The market-based approaches to tackling climate change developed under Kyoto included the Clean Development Mechanism (CDM), International Emissions Trading (IET), and Joint Implementation (JI). The CDM became the hub for compliance carbon market activity: nearly

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8000 projects operated in over 100 countries, leading to the generation of over 1.9 billion certified emissions reductions, or CERs (Michaelowa et al., 2019). Under the CDM, countries were split into Annex I and non-Annex I countries based on their relative economic development. Annex I countries could then transfer funds to non-Annex I countries (i.e., developing countries) in order to support projects that reduced or avoided carbon emissions and supported the sustainable development of the host country (*CDM: CDM-Home*, n.d.). A total of 38 Annex I countries committed to limiting their emissions (not including the United States, which did not ratify the Kyoto Protocol), and therefore participated in trading of CERs in order to reach their emissions reductions targets and be in compliance with the Kyoto Protocol (Michaelowa et al., 2019). Eventually, in 2004, CERs developed under the CDM became eligible for use by the private sector for the European Union Emissions Trading System (EU ETS), which at that point was under development and would be introduced in 2005 (Michaelowa et al., 2019). The EU ETS started as a way for the EU to work toward its Kyoto target—an 8% reduction in greenhouse gas emissions relative to a 1990 baseline—and would go on to become the largest compliance carbon market in the world (*EU Emissions Trading System (EU ETS), n.d.*).

As the CDM gained popularity and became an established compliance market in the mid to late 2000s, an increasing number of companies and governments began participating in parallel voluntary mechanisms (Benessaiah, 2012). In 2003, the World Wildlife Fund (WWF) partnered with other international non-governmental organizations (INGOs) to develop a voluntary carbon offset program and registry called the Gold-Standard, with the mission to maximize the impact of carbon projects and create value for project developers around the world (Gold Standard, n.d.). Then in 2007, businesses and environmental leaders formed the Verified Carbon Standard, or Verra, which today maintains the most active VCM registry, with nearly 1 billion credits issued to date (Thiel, n.d.). These standards sought to build integrity and quality into the carbon markets, while maintaining user friendly but secure registries. The blossoming VCM market and the incumbent CDM framework were already charting the path toward a more functional, transparent market.

The CDM, however, started to lose some of its relevance as the future of the Kyoto Protocol became unclear in the late 2000s. The design of the Kyoto Protocol used commitment periods to reevaluate targets and take stock of the market. As the 2012 commitment period came to a close, activity slowed down in CDM markets and increasingly moved into the voluntary carbon market (Benessaiah, 2012). During this period, the VCM saw a 34% increase in traded carbon credits from 2009 to 2010 (Benessaiah, 2012). Furthermore, in 2010 trade volume in the VCM surpassed the CDM for the first time (Benessaiah, 2012). As the VCM markets developed, the United Nations Framework Convention on Climate Change (UNFCCC) introduced the Reducing Emissions from Deforestation and forest Degradation, or REDD+, framework starting in 2005. REDD + brought land-based voluntary carbon projects to the center of voluntary carbon conversations globally.

In 2015, parties to the annual United Nations Climate Change Conference adopted the Paris Agreement during COP21. The historic agreement called for a goal of limiting global warming to "well below 2 °C″ compared to pre-industrial levels. The VCM, with its network of registries, standards, and frameworks, laid the groundwork toward a developed secondary market for VCM units (UNFCCC, 2015). This burgeoning secondary market now includes financial derivatives products that play important roles in supporting the growth of the market.

2.1. The Paris Agreement

The Paris Agreement, introduced in 2015 at COP21 in Paris, France, is the latest agreement that outlines the goals of international cooperation. With respect to the VCM, the Paris Agreement determines how the market will function and which carbon mechanisms are eligible for inclusion under the agreement. The Paris rulebook, which negotiators completed at COP26 in Glasgow, Scotland, communicates how each country determines and updates their own emissions targets (Nationally Determined Contributions, or NDCs), and the transparency safeguards that parties must follow when reporting their financial contributions, emissions baselines, and progress toward emissions reductions targets (World Resources Institute et al., 2019).

The Paris Agreement also explicitly recognizes carbon markets as a central tool to help countries cooperate to reduce emissions. The agreement calls for the trading of Internationally Transferred Mitigation Outcomes (ITMOs) that must also promote sustainable development. The scale and complexity of the Paris Agreement requires that special care be taken to avoid the double counting of credits as they are traded between entities – this would occur when a private developer would sell carbon units to a buyer in a third country while the government of the project host country would claim that same carbon outcome towards its NDC. The Article 6 negotiations that formalized the Paris Agreement rulebook during COP26 in Glasgow, United Kingdom, introduced the concept of “Corresponding Adjustments” (CAs) to deal with this carbon accounting issue (Fattouh & Maino, 2022). The Corresponding Adjustment mechanism requires that a project's host country makes an accounting adjustment to its NDC when offset credits are transferred internationally – thereby making sure that any carbon emission reduction is only counted once (Zwick, 2021). As parties to the Paris Agreement finalize the processes for implementing Corresponding Adjustments, centralized marketplaces will continue to provide data and transparency to help governments and corporates alike avoid double counting.

The relevance of the VCM continues to increase, and in the runup to and aftermath of the 2021 COP26 climate negotiations major private sector organizations made significant net-zero carbon commitments that involved using offsets to achieve targets (Mountford et al., 2021). Even before COP26, over 1500 companies had put in place net-zero commitments to reduce or offset over 3.5 gigatons of annual greenhouse gas emissions (Streck, 2021). Many market observers now believe

that for major polluters to realistically commit to net-zero carbon goals over the next one to two decades, a healthy, trusted, well-functioning VCM must be in place to meet the growing demand for carbon offsets.

3. Market ecosystem

The VCM ecosystem consists of a variety of public and private organizations, including carbon offset programs (also known as registries or standards), project developers, and carbon buyers, which are often corporate buyers but also include entities such as cities, endowments, and universities. These organizations are all connected via retail providers, financial institutions and brokers that transact carbon credits either through bilateral agreements or participation in exchanges that operate both in spot/cash markets as well as, more recently, futures markets (Bose et al., 2021).

Registries and standards organizations provide a crucial function as they define the monitoring, reporting, and verification (MRV) requirements for each project type. The most established programs are the Verified Carbon Standard (VCS), Climate Action Reserve (CAR), American Carbon Registry (ACR), and Gold Standard. Each registry has its own rules or standards that dictate how project developers originate their carbon credits through methodological approaches to carbon reduction, avoidance, and removal. Once a registry verifies that an offset project meets its criteria, the registry will issue credits against the project activity that quantify the change from the baseline scenario to the low-carbon project scenario – with each offset credit representing one ton of CO2 reduced, avoided, or removed. Those certificates (also called "offsets" or simply "credits") become transferable instruments maintained on each respective registry. Once a buyer purchases a certificate, they can choose to claim the underlying carbon outcome towards mitigating or neutralizing their carbon budget by "retiring" the offset – thereby making sure no one else can claim the carbon outcome the certificate represents.

Retail providers play a large role in structuring deals and matching project developers and corporate buyers – they are able to provide advisory services and market expertise for buyers that are not as familiar with the VCM structure. Those firms can also hold inventory of offset credits. While some (large) corporate buyers may choose to produce carbon offsets directly from project developers, retail providers and brokers play an important role in the market. In particular, they are often able to help financing projects at an early stage by committing to buy a minimum amount at a fixed price agreed today. The varied nature of projects available to buyers makes pricing challenging and raises the issue of information asymmetry. Project developers and buyers would need to keep track of competitors' bilateral agreements to see what a "fair" market price would be. Additionally, they would need to consider any differences between competitors' projects and their projects vis-a-vis project type or technology, geography, and methodology in order to understand the value of the credits under negotiation. This system of ad-hoc, bilateral carbon sales creates an opaque trading environment in which negotiations are protracted and incur a high level of effort and transaction costs for both parties. The burden of slow negotiations and high transaction costs falls disproportionately on project developers that depend on financing and are eager to land a deal, while well-capitalized buyers can take time to build up their carbon project inventory. In some instances, uninformed project developers have received a lower price for the credits than other comparable projects, which would lead to public criticism and claims of corporate greed or unethical behavior that would damage the entire industry's reputation (Hodgson, 2022). As the VCM continued to develop, the opacity and ongoing evolution of the market indirectly led to quality issues with certain credits. Projects undertaken in previous years might have been verified under newly developed, less rigorous monitoring and verification requirements, and the early iterations of methodologies may have been untested (Carbon Pulse, 2022). Without clear information on which projects carried a higher level or rigor or integrity, buyers ended up purchasing credits that might not actually represent any reduction or avoidance in CO2 emissions (Pearce, 2021). Savvy traders could find ways to sell these credits, generating revenue on no or little actual emissions impact. As time went on, the credits that carried an older vintage continued to slip into purchase agreements, and when discovered, would sometimes lead to public criticism and accusations of a fraudulent, poorly functioning market (Pearce, 2021).

Exchanges have emerged to facilitate and optimize the price discovery process in the VCM away from the OTC bilateral market. By establishing standard trading terms, exchanges are also seeking to improve transparency in the VCM space. CBL Xpansiv is the largest spot exchange for offset credits. Carbon-offset volume transacted on its platform exceeded 120 million metric tons of CO2 equivalent (mt CO2e), a 288% increase from 2020 levels (Spilker, 2022). Other exchanges active in spot markets include Climate Impact X and AirCarbon Exchange in Singapore as well as the Carbon Trade Exchange (Sydney and London). Futures exchanges have now also entered the space, providing forward price discovery and liquidity in the secondary market trading of offset credits.

4. Role of derivatives in the marketplace

There is an established marketplace for carbon derivatives in compliance markets. Europe is home to the world's most developed carbon market, the EU Emissions Trading System (EU ETS). According to Refinitiv, the EU ETS accounted for 12.2bn MT out of 15.8bn MT carbon traded, or EUR 683bn out of EUR 759bn (90% of the total) in 2021 (*Refinitiv Carbon Research*, n.d.). Trading of EU ETS allowances via futures is well established, with multiple venues offering futures and options contracts. Compliance products are well suited for futures markets since they are standardized instruments – each emission certificate is perfectly fungible with each other and can therefore easily be traded on centralized exchanges as well as OTC trading venues. Beyond the EU ETS, derivatives

markets have emerged for the UK ETS following Brexit as well as RGGI[1] and California Carbon allowances in North America.

Derivatives on VCM products have taken a longer time to emerge. Unlike in compliance markets, where one central entity issues the compliance instrument via an auction or allocation process, voluntary markets depend on carbon projects that are certified by registries. Once certified, these carbon offset credits become freely tradable, but they have unique characteristics that are defined by the type of project, its geography, the vintage year, the certifying entity, and its certifying process as well as associated co-benefits. This spectrum of unique characteristics makes standardization all the more challenging, something that is reflected in a wide distribution of prices in the VCM space (Bose et al., 2021).

In 2021, CME Group launched a suite of VCM futures in partnership with CBL/Xpansiv, the leading spot exchange for offset credits (CME Group, n.d.). The first VCM futures contract to launch, the CBL Global Emissions Offset (GEO) futures is a physically deliverable contract and allows for the delivery of CORSIA-eligible voluntary carbon offsets from VCS, ACR, and CAR. CORSIA refers to Carbon Offsetting and Reduction Scheme for International Aviation, a market-based mechanism adopted by the International Civil Aviation Organization (ICAO), a specialized UN agency, to meet an ambitious goal of carbon neutral growth from the international aviation industry starting in 2020.

Shortly after, and recognizing the vital role that Agriculture, Forestry and Other Land Use (AFOLU) is playing in the VCM, CME Group also launched a CBL Nature-based Global Emissions Offset (N-GEO) futures contract that allowed the delivery of offset credits from Verra AFOLU projects. To capture additional co-benefits in each carbon credit, CME Group's N-GEO contract terms stipulate that any deliverable contract also needs to conform to one of Verra's Climate, Community, and Biodiversity (CCB) standards. Building on the learnings of the VCM over the previous decade, exchanges like CBL Xpansiv and CME Group are standardizing carbon contracts with built-in certifications like CCB to ensure that the derivatives markets capture the co-benefits and sustainable development goals originally enshrined in the CDM and frameworks like REDD+.

Finally, in 2022, CME Group launched a CBL Core GEO Global Emissions Offset contract (C-GEO). CBL C-GEO futures are standardized instruments for high-quality voluntary emissions offsets that are intended to align with the Core Carbon Principles (CCPs), an emerging set of transparent and consistent standards that aim to build quality and integrity into the supply of carbon credits. The Integrity Council for the Voluntary Carbon Market (ICVCM), a private sector-led initiative, oversees the Core Carbon Principles and is seeking to address quality and transparency issues in the VCM with the goal of building trust and scaling the market (ICVCM, n.d.). The ICVCM also define which standards and methodologies are eligible to be aligned with the CCPs (ICVCM, n.d.). Most recently, CME Group developed "Trailing" futures, which follow the criteria set for CBL N-GEO and CBL C-GEO but deliver older vintage years of available offset credits. Because the vintage eligibility window for CBL N-GEO and CBL C-GEO contracts moves forward each passing year, market participants can be sure that they are transacting credits of recent vintage when trading the main benchmark contracts. The availability of "Trailing" CBL N-GEO and CBL C-GEO contracts helps maintain liquidity and volume in offset credits that became ineligible in the main benchmark contracts.

Other exchanges have also launched futures that target the VCM space, with the InterContinental Exchange (ICE) launching nature-based VCM futures and Nodal/EEX providing Global Emission Reduction futures, the latter encompassing both nature- and technology-based solutions in its delivery basket.

At CME Group, the three available contracts have found quick adoption by the marketplace, with both volume and open interest continuing to grow since their launch. One pattern stands out in particular: futures contracts are routinely used for physical delivery. More than 5.9 m MT (5900 contract equivalents) were physically settled upon expiry of the December 2021 contract.[2] This means that the contracts are used to make and/or take delivery of offset credits, thereby providing a direct link to the physical spot market for offset credits. This physical delivery activity demonstrates that the contracts are filling a useful role and provide a link between physical and derivatives markets – connecting carbon projects developed across most of the developing world to financial centers in Chicago, London, and New York.

The development of futures markets around the VCM is in an early phase, but the development already reflects typical patterns of commodity markets as they grow in size and attract more capital and intermediaries (Roman, 2022). Early market structure usually consists of long-term, fixed price deals between a limited number of sellers and buyers. As the market grows and the number of participating firms expands, buyers and sellers might reconsider fixed-price deals and pivot towards floating price indexation – if expectations of future prices are too divergent, and renegotiating fixed price deals is too costly, index adoption offers an elegant solution. Typically, index prices are derived from spot market activity and thus represent the cost at which a marginal commodity unit could be transacted today. For derivatives markets to emerge, there needs to be available spot reference prices as well as an appetite by issuers, intermediaries, and purchasers to use those indices and/or futures in their physical supply contracts. As market acceptance grows, future issuance volumes can be tied to that index, or to futures market prices that track this index. As a next step in the evolution of those markets, we are seeing the emergence of exchange traded funds that hold futures

[1] RGGI stands for the Regional Greenhouse Gas Initiative, a cap-and-trade program covering several North-Eastern States in the U.S.

[2] Carbon budgets are usually managed on a calendar year basis. By market convention, December futures tend to be the most widely traded contract since they expire at the end of the year. This pattern is observable on both compliance and VCM futures markets.

contracts. The means that ETF owners are indirectly holding a long or short position in underlying offset credits. More recently, blockchain tokenization, in which offset credits are tokenized and traded on blockchain applications, has become a new route to (the secondary) market for offset credits. This development is not free of criticism and certain registries have already voiced their opposition to blockchain tokenization as it is currently practiced (Thiel, 2022). The main criticism of tokenization is that credits are retired prior to tokenization, and some argue that this means that the token has zero intrinsic value (Zombies on the Blockchain – CarbonPlan, n.d.).

The emergence of reference prices, and derivatives tied to those prices, is a significant development for the VCM sector. Price transparency in the sector has been a challenge in the past, with accusations that project developers – often in the global south – were left shortchanged by savvy intermediaries able to procure low-cost offset credits that they then resell at high margins (De Haldevang, n.d.). The emergence of futures contracts, for which exchanges make settlement prices available to the public, means that project developers now have a means to see where the market values benchmark carbon offsets for both prompt delivery and future delivery periods. This price transparency should be highly valuable to project developers: just like the farmer in Iowa that looks up CBOT grain prices, or the smallholder in Cote d'Ivoire tracking global cocoa reference prices, project developers now have a price reference they can use to budget, negotiate, and execute carbon offset projects.

Standardization in the derivatives world means that those reference prices will not be exactly equivalent to any unique project, but they can form the basis of price negotiations, with project specific qualities captured by premiums or discounts attached to reference prices. In listing different futures contracts, CME Group and other exchanges are aiming to establish different reference prices to cover separate sectors, such as the CORSIA-aligned market (covered by the C-GEO product), nature-based solutions (NbS) and natural climate solutions (covered by N-GEO), and IVSCM-aligned projects (covered by the C-GEO product). The VCM is going down the well-trodden path of futurization, with benchmark prices emerging that provide reference values for projects. This trajectory mirrors the development of the oil derivatives markets, with physically deliverable futures contracts on WTI and Brent setting the global reference prices in crude oil and regional quality differences captured in a premium/discount versus those global reference prices.

5. Summary of trading activity of CME group VCM products

In their short trading history, prices of VCM carbon futures have shown a high degree of volatility, with the price of the nearby futures contract increasing fourfold in the span of nine months in the case of CBL GEO futures and more than doubling in the case of CBL N-GEO futures. Prices reached a peak around late 2021 and the first quarter of 2022. The price increases were attributed to optimism around COP26, the U.N.'s Climate Change Conference which suggested a large role for voluntary carbon markets in efforts to reach Net Zero. With the beginning of the war in the Ukraine, sentiment began to shift, and prices fell back somewhat into the summer months of 2022. The initial shock may have been the Russian invasion, but wider worries about economic growth and soaring inflation also clouded the short-to medium-term outlook. In July 2022, nature-based credits (in the form of CBL N-GEO) traded at a premium to tech-based solutions (captured by CBL Core-Geo and CBL GEO futures), indicating higher demand for these types of projects (See Chart 1).

Trading activity in deferred contract months shows that the market for VCM Futures – here representatively N-GEO futures, which are the most widely traded VCM instrument on the CME Group exchange – shows a contango structure. Contango is a market environment in which futures contracts with a delivery date further into the future trade at a premium to nearby contracts. A contango market typically shows market participants' expectations that prices will rise from current levels due to increased demand, reduced supply, or a combination of both. This also means that despite spot price weakness – as witnessed by falling nearby prices from the highs recorded at the beginning of the year – market participants expected prices to rise from the levels seen in July 2022 into future delivery years. In addition, the fact that trading activity extends more than 3 ½ years from the current date is a significant endorsement of VCM derivatives markets and the role they can play.[3] Typically, trading activity in futures contracts – and relatively new products especially – is heavily concentrated in nearby months. For a market that used to trade mostly in the form of fixed price deals or spot OTC transactions, the emergence of forward liquidity in the form of standardized contracts is a true innovation. This is especially relevant to the VCM space as project horizons often extend decades into the future, meaning that issuances expected to be originated in coming years can now be valued at prices reflecting forward market expectations (See Chart 2).

Trading activity on the contracts shows that the nature-based futures have been the most popular to date, accounting for about 70% of total trading activity on CME Group, followed by CBL GEO futures at 25% and finally CBL C-GEO futures at 5%. Market participants are still awaiting the finalized Core Carbon Principles (CCP) from the ICVCM. Higher certainty from the ICVCM could in turn increase adoption of carbon offsets and the use of futures markets like CBL C-GEO futures for risk management and price discovery.

Open interest, the number of contracts held open by market participants, is one of the key indicators to assess adoption of a futures contract. Open interest has been on a steady incline and peaked above 20,000 contracts in late May 2022. Mirroring trading activity, most open interest is held in nature-based contracts (approx. 80% of total) (See Chart 3).

[3] At the time of writing, the nearby N-GEO futures contract is the July-2022 contract month. Trading activity extends out to December-2025 contract month.

Chart 1. VCM Futures prices on CME Group, Prices in USD of C-GEO, GEO, and N-GEO from March 2021 to June 2022. Source: CME Group, data as of July 12, 2022.

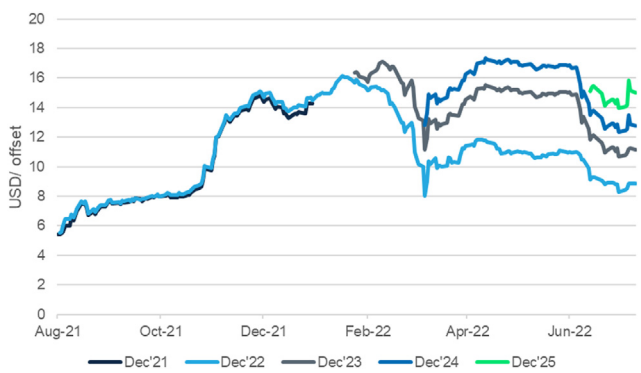

Chart 2. CBL N-GEO term structure. N-GEO term structures for Dec'21, Dec'22, Dec'23, Dec'24, and Dec'25, with prices in USD from August 2021 to June 2022. Source: CME Group, data as of July 12, 2022.

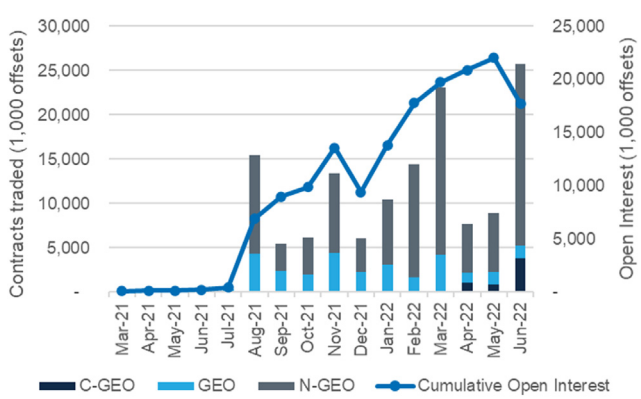

Chart 3. Volume and Trading activity. Volume and trading activity of C-GEO, GEO, and N-GEO products, showing number of contracts traded, open interest, and cumulative open interest form March 2021 to June 2022. Source: CME Group, data as of July 12, 2022.

As mentioned above, market participants have made wide use of the physical delivery feature of the CME Group VCM contracts, with more than 10,000 contracts (10 m offsets) physically delivered via the regular contract expiry process. For physical contracts, the delivery process provides the mechanism for convergence between spot and futures prices. The fact that a significant number of contracts were physically settled upon expiry is a strong endorsement of VCM derivatives and

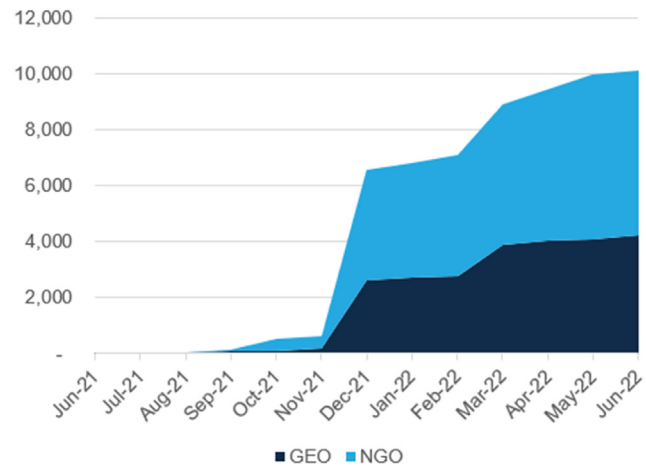

Chart 4. Physical delivery. Cumulative physical deliveries of GEO and N-GEO from June 2021 to June 2022. Source: CME Group, data as of July 12, 2022.

how they can be used to make or take delivery of individual project credits using standardized contracts (See Chart 4).

6. Literature review

Voluntary carbon markets (VCM) are a relatively new field of academic research. The vast majority of financial research written on carbon derivatives focuses on the established EU-ETS market, for which price and other market data can be retrieved going as far back as 2005. In comparison, research on VCM derivatives is still limited by the relatively short trading history of those derivatives contracts – less than two years in the case of CME Group VCM futures.

Published academic research on the VCM focuses mostly on policy and governance issues rather than the emerging derivatives space. Our review identifies challenges with market transparency and integrity both at the macro level (e.g., issues around Article 6 of the Paris Agreement, CAs, and analyses of competing standards) and at the project level (e.g., investigations of additionality, permanence, and leakage issues; project development costs; etc.). We reviewed published literature and, where applicable, considered how these issues may impact carbon derivatives, particularly with respect to transparency, standardization, and price discovery. A portion of the literature discusses from a higher vantage the characteristics of derivatives in carbon markets, especially the value that exchange-traded futures may bring to both compliance and voluntary markets.

Ahonen et al. (2022) look at the governance models underlying the historical evolution of global carbon market. They note that the CDM was a top-down driven model while the new developments under the Paris Agreement and COP26 are characterized by bottom-up governance systems emerging and competing with each other. They offer that the main building block in a functioning market for carbon credits is the legitimacy of the governing institution issuing the credits. Shortcomings of the CDM led to the emergence of private registries such as Verra and the Gold Standard which brought diversification and competing governance systems to the market at the

cost of increasing fragmentation. They do note, however, that there are attempts to achieve "meta-governance" in the VCM space through initiatives such as the ICVCM. Private registries are involved in these efforts by preparing to implement attributes to distinguish Article 6 or CORSIA eligible credits. In the context of derivatives, "meta-governance" initiatives resulting in better aligned standards in VCM spot markets could pave the way for further market growth in derivatives.

Kreibich and Hermwille (2021) focuses on the perceived mismatch between the supply of offset credits and growing demand for carbon offsets from the private sector. They caution that carbon offsetting using VCM products should not provide an "easy" way out for the private sector. In other words, offsetting should be accompanied by credible efforts to reach net zero. The way the VCM functions under the Paris Agreement is also under scrutiny, in particular the way demand from the private sector may conflict with NDCs when the same ton of carbon is claimed by a private corporate as well as the country that hosted the carbon project. The authors believe that any future unit bought and sold without a CA linkage to the accounting structure of the Paris Agreement may set off a race to the bottom by competing offset providers. The authors conclude that VCM offsets with appropriate CAs are the only feasible solution to build integrity into the market and satisfy skeptical observers. The authors note that there are technical challenges to that, but that the main issues are of a political nature. While the application of CAs is still under discussion, differing approaches by standard setters could result in separate liquidity pools for trading and hedging risk, which could consequently hinder market growth in the derivative space.

Similarly, studies note that the post Paris Agreement environment has resulted in a proliferation of regulatory and supervisory efforts. They are optimistic that the Article 6 rulebook is creating conditions for effective and robust international carbon markets to strive yet also caution that private actors need to be mindful of implications of CAs, governance frameworks, and financial and reputational risk of investing in countries that may not be able to meet their NDC (Fattouh & Maino, 2022; Kreibich & Hermwille, 2021). These papers conclude that Article 6 of the Paris Agreement has the potential to encourage convergence across VCM and compliance markets (Fattouh & Maino, 2022; Kreibich & Hermwille, 2021). Any rapprochement between VCM and compliance markets would be a significant development across spot and derivatives markets, with possible liquidity spillover effects from the mature compliance markets into the nascent VCM space.

Streck (2021) finds that when the VCM complements government action, the aforementioned risks around the VCM can be managed. The author notes that VCM has grown as the response to the twin forces of government inaction on climate change combined with public pressure on corporations to do more to manage their emissions. Streck also places VCM in the context of mostly western corporations funding projects in developing countries and cautions that greater volumes of offset credits cannot come at the cost of harming local communities – something that registries are seeking to address through the inclusion of metrics to track co-benefits. The author also sees the scope for public-private partnerships in the development of VCM, with governments providing the environment in which private companies have the necessary governance framework to invest in local projects. Finally, the author also illustrates how VCMs can promote promising new technology to remove carbon. CME Group N-GEO futures as well as other similar nature-based futures already stipulate that the underlying projects need to include additional labels that address the provision of co-benefits, such as Verra's Climate, Community, and Biodiversity (CCB) label. The characteristics of the underlying carbon futures assets reflect the strong interest of market participants in ensuring high-quality, high-integrity projects.

Focusing on issues at project levels, Si Chen et al. (2021) conducted an empirical study of market participants in the VCM. They observe that transparency and quality are key issues for participants, yet the high cost of top-down protocols and compliance needed to improve quality also runs the risk of limiting the growth of the marketplace. The authors interviewed more than 30 stakeholders across the VCM supply chain who stated that quality, transparency, and costs (in that order) are the most pressing challenges. According to the interviewees, the primary challenge around quality is the question of additionality – the certainty whether a carbon outcome achieved by a project was "additional" to what would have happened without the project. The lack of transparency is understood as the logical consequence of a highly fragmented marketplace. The issue of high project development costs is partly due to high overhead costs for project monitoring, reporting, and verification (MRV), which impacts the ability of developers to raise funds with offsets. The authors add that unscrupulous buyers and sellers could severely jeopardize the functioning of the market. A buyer seeking the lowest quality offset to "check a box" and a project developer cutting corners during the MRV process would both be detrimental to credit quality and integrity, ultimately undercutting trust in the market. The authors also observe that higher quality requirements result in higher verification costs. They also add that programs like CORSIA help form industry standards (for example, in the airline industry) and could be adopted by other industry sectors. One caveat to the authors' analyses is the small sample size for their industry survey and that the interviewee selection process included, according to the authors, some unavoidable selection bias. The emergence of robust price signals from the derivatives marketplace could help address transparency issues: while project specific features, including unavoidable MRV costs, will always need to be priced in, a public reference price for standardized contracts in the derivatives space should at least provide more clarity and set a benchmark against which a premium or discount can be applied.

As mentioned above, nature-based solutions have found high interest from the market, something which is also mirrored in market activity in nascent derivatives markets with N-GEO

contracts attracting the most volume to date. In various studies, researchers have found that NbS carbon emission reductions and the corresponding number of offset credits generated may have been overstated in the past. This applies to regions and markets as diverse as REDD + projects in the Brazilian Amazon (West et al., 2020) or offsets eligible under the California Cap and Trade program (Ruseva et al., 2017). While the issues are not directly related to the functionality of the VCM derivatives market, reputational risk around VCM offsets could hinder market growth. Spot markets provided by exchanges such as CBL/Xpansiv can help to increase trust in the characteristics of underlying assets by requiring that the carbon assets meet high verification criteria provided by the major standards such as Verra and Gold Standard. The competing and dynamic nature of VCM registries also leads to standards developing new labeling criteria to capture co-benefits as well as evolving monitoring and verification methodologies over time in order to increase rigor. Streck states that leading standards in the VCM market can provide assurances around credit quality, in particular with respect to co-benefits related to sustainable development of local communities.

The available literature discusses how increased transparency and market efficiency are characteristic of carbon futures exchanges addressing the compliance market, but that these effects could also apply to the voluntary market (Chen Y., 2021). Futures market activity can be described as "a comprehensive indicator of both supply and demand parties' expectations" of future price and supply. Therefore, price data on carbon futures exchanges is "consistent, open, prescient" and can lead to increased transparency in the market and efficient allocation of resources. Indeed, a "full-fledged" carbon market is "one that aligns and combines carbon spots with carbon forwards" and thus a commodity market for carbon "is characteristic of alignment and unity" (Chen Y., 2021).

Analyses show that carbon futures help a market "work as efficiently as possible" as price signals are communicated through standardized futures contracts traded over a transparent exchange (Mensah, 2014). This increased transparency can limit "excessive price volatility" and contribute to a more efficient market (Mensah, 2014). Because futures are exchange-traded, they provide crucial transparency that may be lacking in the opaquer OTC market – where clearinghouses and timely provision of trade data are lacking (Mensah, 2014; Newell et al., 2014; Schneck & Monast, 2011). The mitigation of counterparty risk that comes with exchange-cleared trading increases efficiency and transparency, as does participants' equal distribution of and access to market data (Schneck & Monast, 2011). While market data for recently developed VCM futures and other derivatives is relatively new, over time we may see increased research on market performance that will test the existing literature's claims that well-developed futures exchanges increase overall transparency and thus contribute to a functional, high-integrity marketplace for (voluntary) carbon markets.

7. Paper contribution, observations, and policy implications

This paper is an effort to explore the nascent stages of exchange-traded voluntary carbon futures and what role they may play in the VCMs. The recent development of carbon derivatives products in financial markets, led by CME Group's launch of the CBL GEO and CBL N-GEO futures, is a sign that the VCM is maturing. While estimates of future market size vary widely–from $50bn (Blaufelder & Levy, 2021) to $180bn (Institute of International Finance, 2021) by 2030–the overarching consensus is that the VCM is poised for significant growth in the years to come. Some of this growth is due to the fact that voluntary carbon markets are playing an increasingly important role on the road to Net Zero. As the market continues to grow, appropriate policies and products will help to ensure an efficiently functioning market. Although the body of literature covering derivatives in the VCMs is relatively small, we have collected the following observations in an effort to lay the groundwork for further research as the market develops.

The emergence of VCM futures markets is a welcome development with potential benefits for the entire carbon supply chain. In particular, it is worth highlighting the additional pricing transparency and standardization that derivatives can bring to the table. In a market where project prices vary according to many characteristics, having benchmark prices available for different sectors will help contract negotiations and promote market efficiency.

Some studies show that currently there is not enough supply to meet future demand, and efficient and fast financing can help bridge this gap (Nowak, 2022). Centrally cleared derivatives such as futures allow capital efficient trading and hedging as well as a significant reduction of counterparty risk, with a well-capitalized clearing house sitting between the long and short side of each derivatives trade (Mensah, 2014; Newell et al., 2014; Schneck & Monast, 2011).

Issues that have created bottlenecks that hinder the efficient functioning of the market to date, such as opaque pricing and lack of standardization, are now being addressed through the price discovery that a derivatives market provides. Efforts to "commodify" VCM products–that is, the standardization of carbon offset credits into tradable benchmark products–is helping to transform credits that vary in quality and integrity into mutually interchangeable commodities. The commodification of carbon credits through standardized contracts, if done well, can help increase forward liquidity, volume, and price transparency to the benefit of carbon developers, buyers, and traders alike. We hope that our observations can inform policy decisions and inspire further research, some suggestions for which we lay out below.

Derivative product providers can support market development by offering standardized solutions and improving market transparency. For strong derivatives markets to emerge, the VCM ecosystem needs to continue developing standard

frameworks and evolving governance models that build trust in the underlying assets. The emergence of forward liquidity will also help project developers and their financiers value projects across a longer project horizon, potentially unlocking more climate finance. Unlocking this finance is critical so that project developers can receive the funding necessary to develop projects and meet the burgeoning demand for credits.

Targeted market education for project developers, intermediaries, and end-users will be crucial to further market growth. This will ensure that increasing numbers of participants are aware of the potential benefits of VCM "commodification" and "futurization". VCM futures contracts need to address a well-defined market segment and have relevancy to the marketplace. Product innovation in the derivatives space, whether in the futures, ETF, or crypto space, should always follow the evolution of the physical market and the way that market standards evolve. Recently, efforts to tokenize credits have generated a lot of excitement, but tokenization has also obscured vintages, leading to projects with old vintages being repackaged to be more marketable, and in some cases even leading to already retired credits resurfacing and being resold (Mulder, 2022). While product providers should not and cannot determine market prices, they should make sure that the products they promote represent the most widely accepted and reputationally-sound offset projects available, which in practice means products that do not trade at highly discounted prices. Any product designed for superannuated and/or already retired credits should be clearly defined to ensure transparency.

The VCM ecosystem consists of various registries, spot and futures trading platforms, ETF providers, price reporting agencies, and other entities. This varied group of stakeholders should, when feasible, continue to coordinate efforts and speak with a common voice. Further market growth will also be helped by regulatory certainty. Industry organizations, like the International Emissions Trading Association, or IETA, and the Integrity Council for Voluntary Carbon Markets, or ICVCM, are crucial in order to promote that common voice and push for the development of commonly accepted market standards. The emergence of the VCM as a futures and derivatives asset class also requires that regulators and standard setting bodies such as the Commodities Futures Trading Commission (CFTC) in the U.S., its international regulatory agency counterparts, and organizations such as ISDA have a strong appreciation of the underlying market. The CTFC is already taking the first steps and convened a workshop on the VCM in June 2022. Furthermore, the Taskforce on Scaling Voluntary Carbon Markets report published by the ICVCM provides an important framework and guideline document to further steer the evolution of the VCM market (Institute of International Finance, 2021). Lastly, market participants will need to pay close attention to the implementation of Article 6 and its implications on the VCM space, especially with regard to Corresponding Adjustments (Fattouh & Maino, 2022).

Our review of existing literature illustrates policy challenges at both the macro and project level. Each of these challenges will ultimately impact the health and functionality of the derivatives market. Assuming that the market continues to grow, future academic research in the field of the VCM may focus on the interaction of compliance and voluntary markets at a granular level, correlations to other asset classes in general, as well as the behavior of different types of traders in the market. The quality of such research, more than anything, will depend on having historical market data in a sufficiently liquid derivatives contract. As VCM derivatives proliferate over time, a rich new field for academic research is likely to surface.

Declaration of competing interest:

There are no conflicts of interest. Both authors work at CME Group. Nick Nugent is the corresponding author.

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